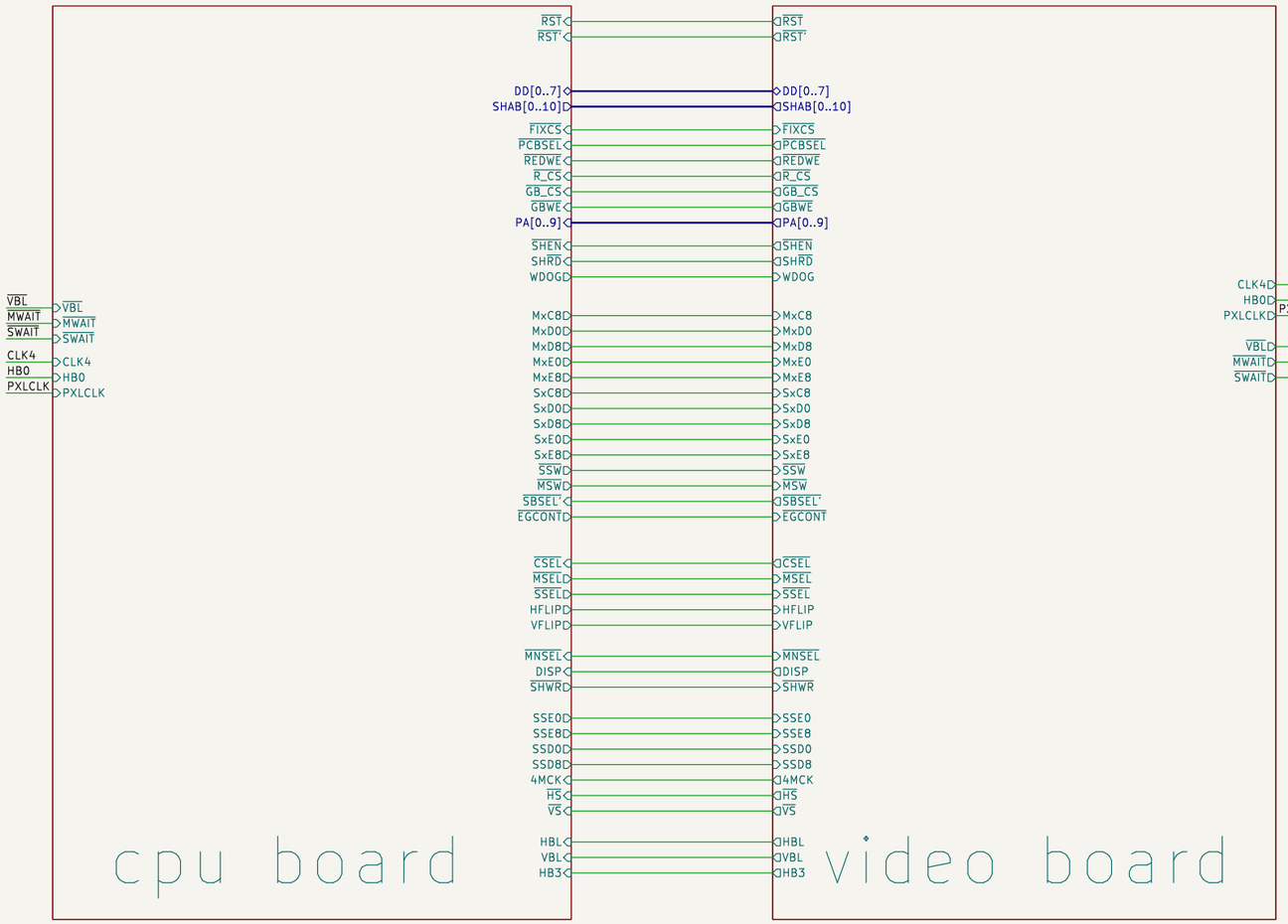


tehkanwch bootleg board

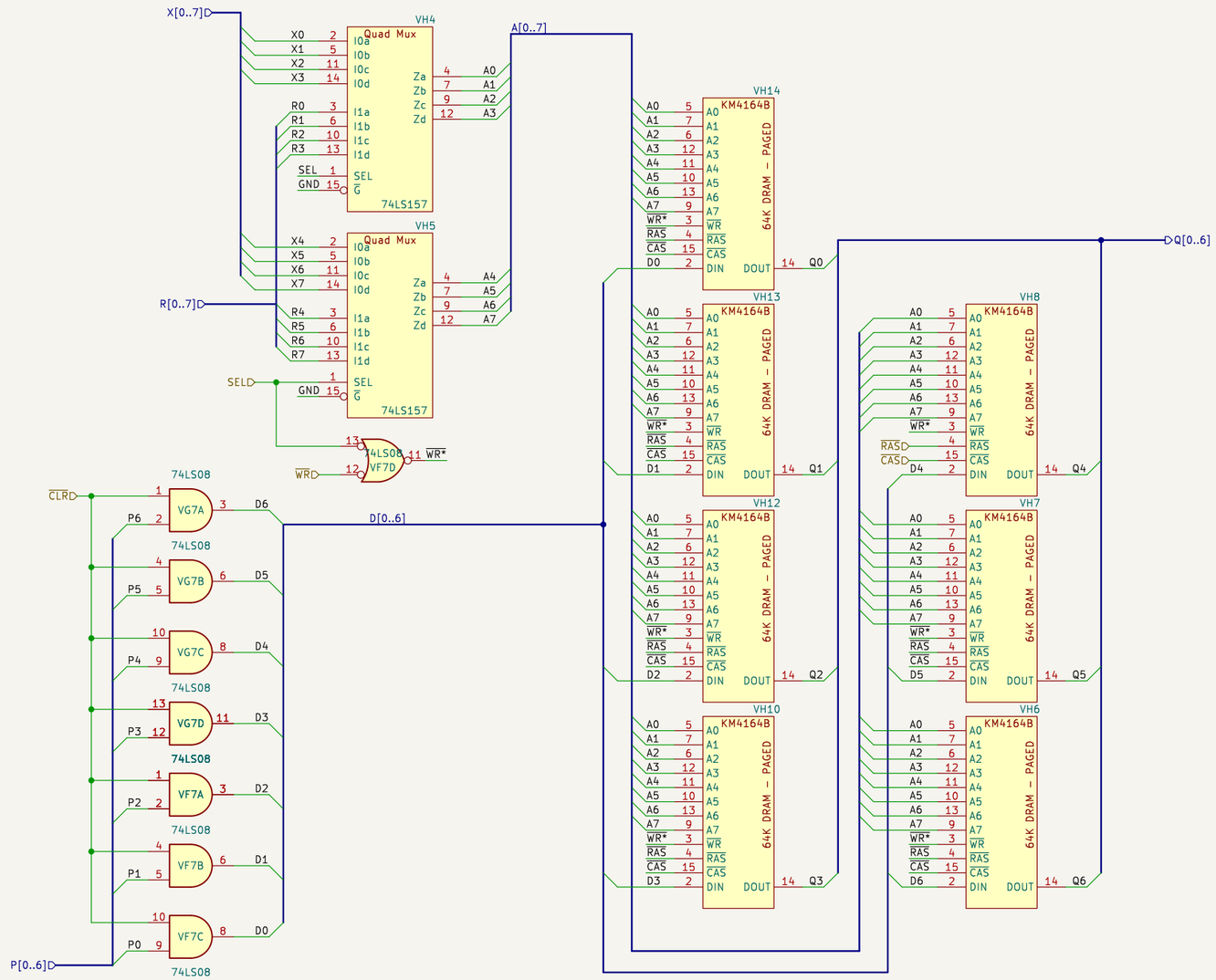
- original PCB has through-hole vias whereas the bootleg has filled vias
- original uses three custom packages from Mitsubishi with logic inside.
  These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
- debug edge connector on video board not present on bootleg
- cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
- different power amplifier for sound without heat sink on the bootleg
- slightly different frequencies for crystal oscillators (cost reduction?)
- device reference markers (silkscreen) on bootleg have reversed order on bootleg

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Sheet: /
File: tehkanwch.kicad_sch

Title: TEHKAN WORLD CUP (tehkanwch bootleg)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/26

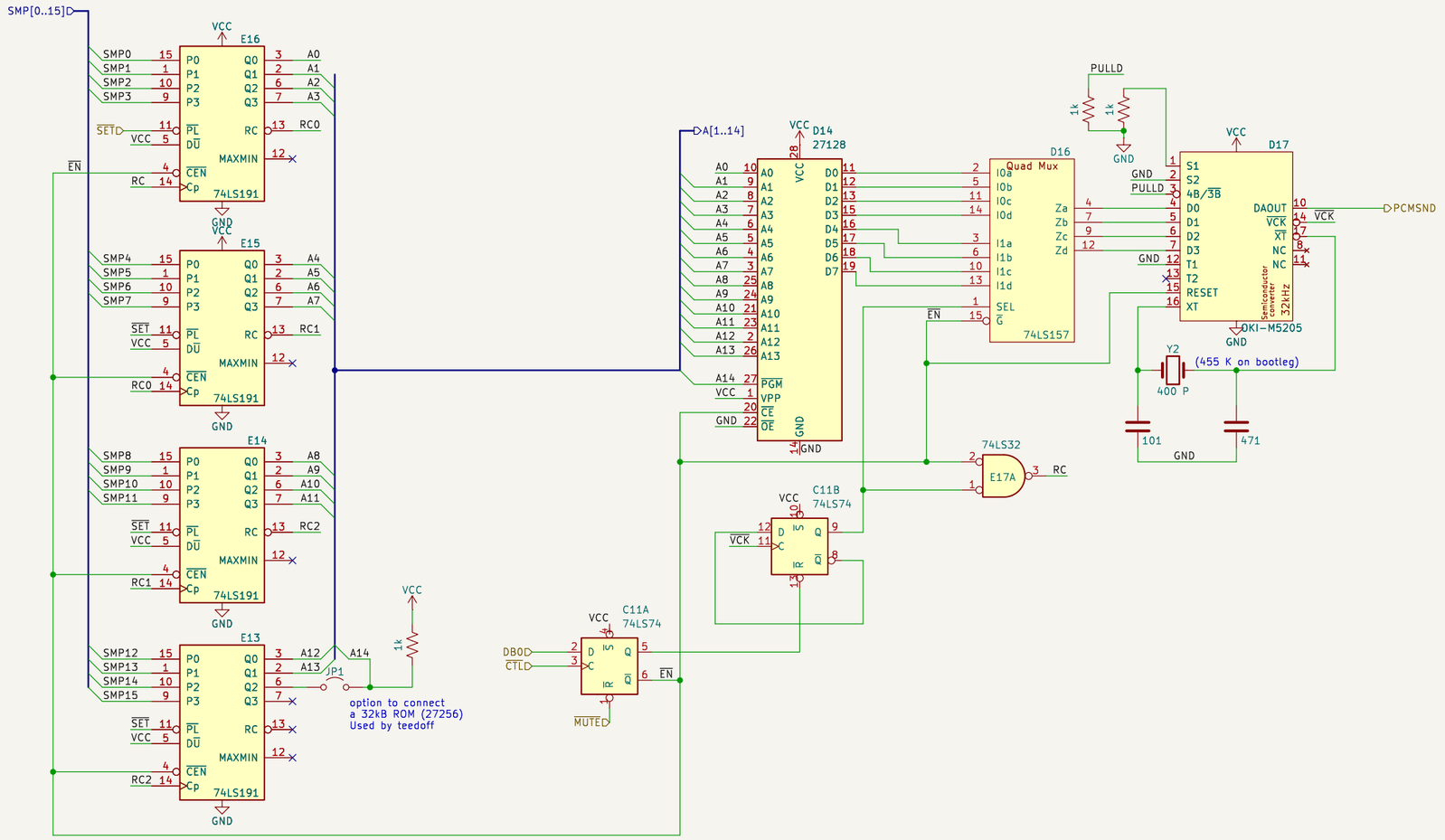


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Sheet: /video board/object/frame buffer B/
File: buffer.kicad_sch

Title: Line Buffer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 2/26

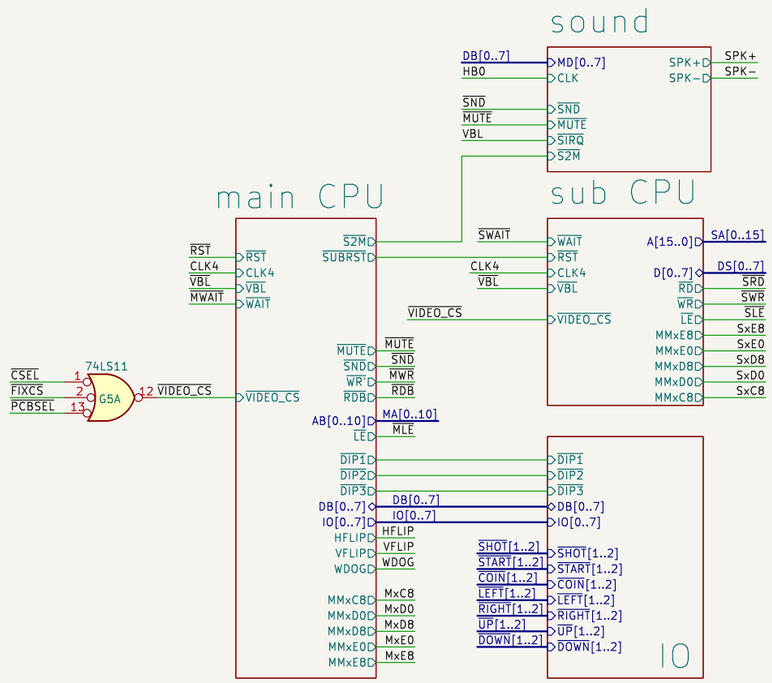


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Sheet: /cpu board/sound/PCM/
File: pcm.kicad_sch

Title: PCM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 5/26

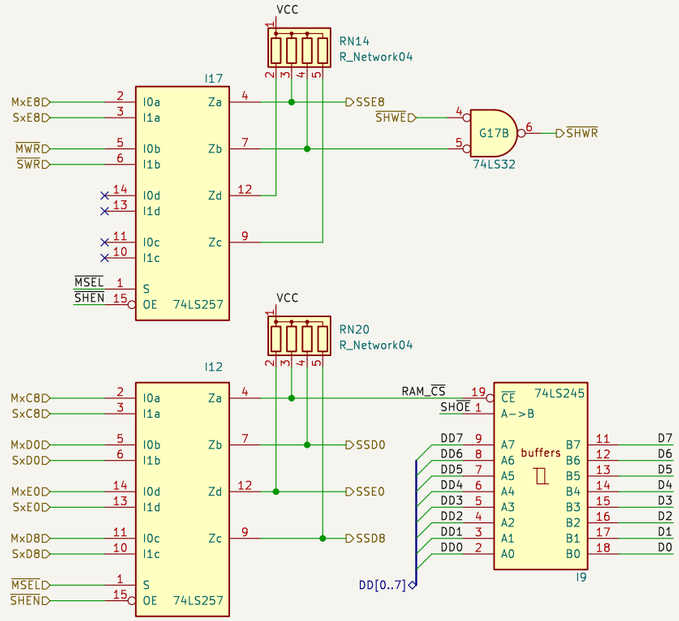


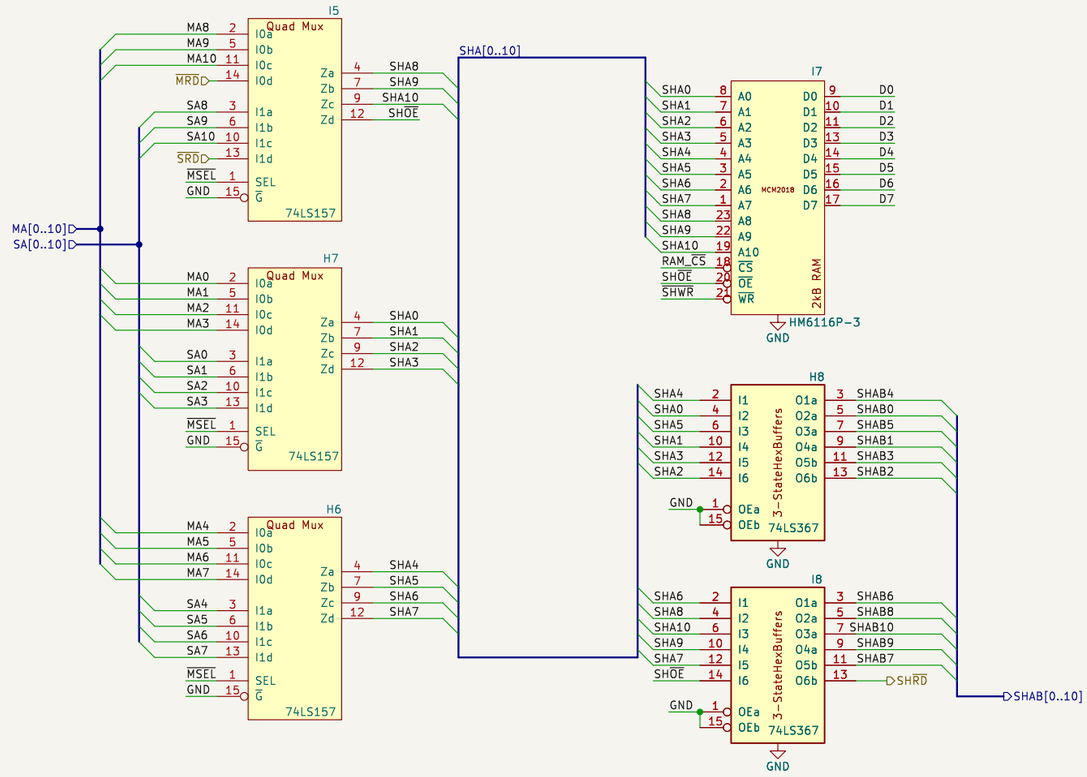
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Sheet: /cpu board/
File: cpu.kicad_sch

Title: CPU board (top board)

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 9/26







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Sheet: /cpu board/shared/
File: shared.kicad_sch

Title: Shared Memory

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 10/26

timing

object

scroll

fix

color mixer

watchdog

address decoder

VJ1

VJ2

connectors
to CPU board

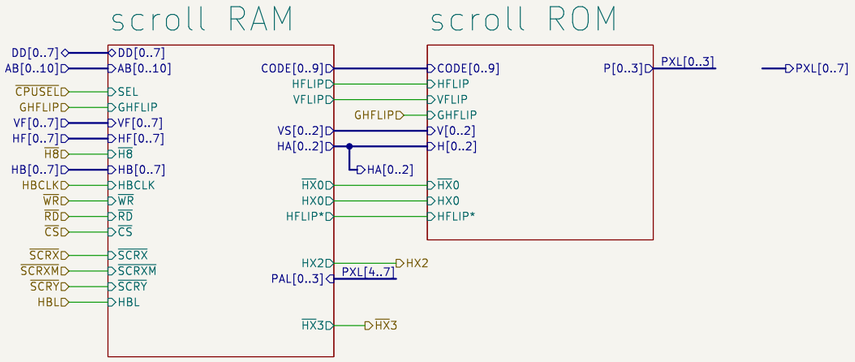
VJ1 at PCB's top right
VJ2 at PCB's bottom right
pins counted from board bottom up,
pins at the top row of the connector
counted as even

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Sheet: /video board/
File: video.kicad_sch

Title: Video board (bottom board)

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 11/26

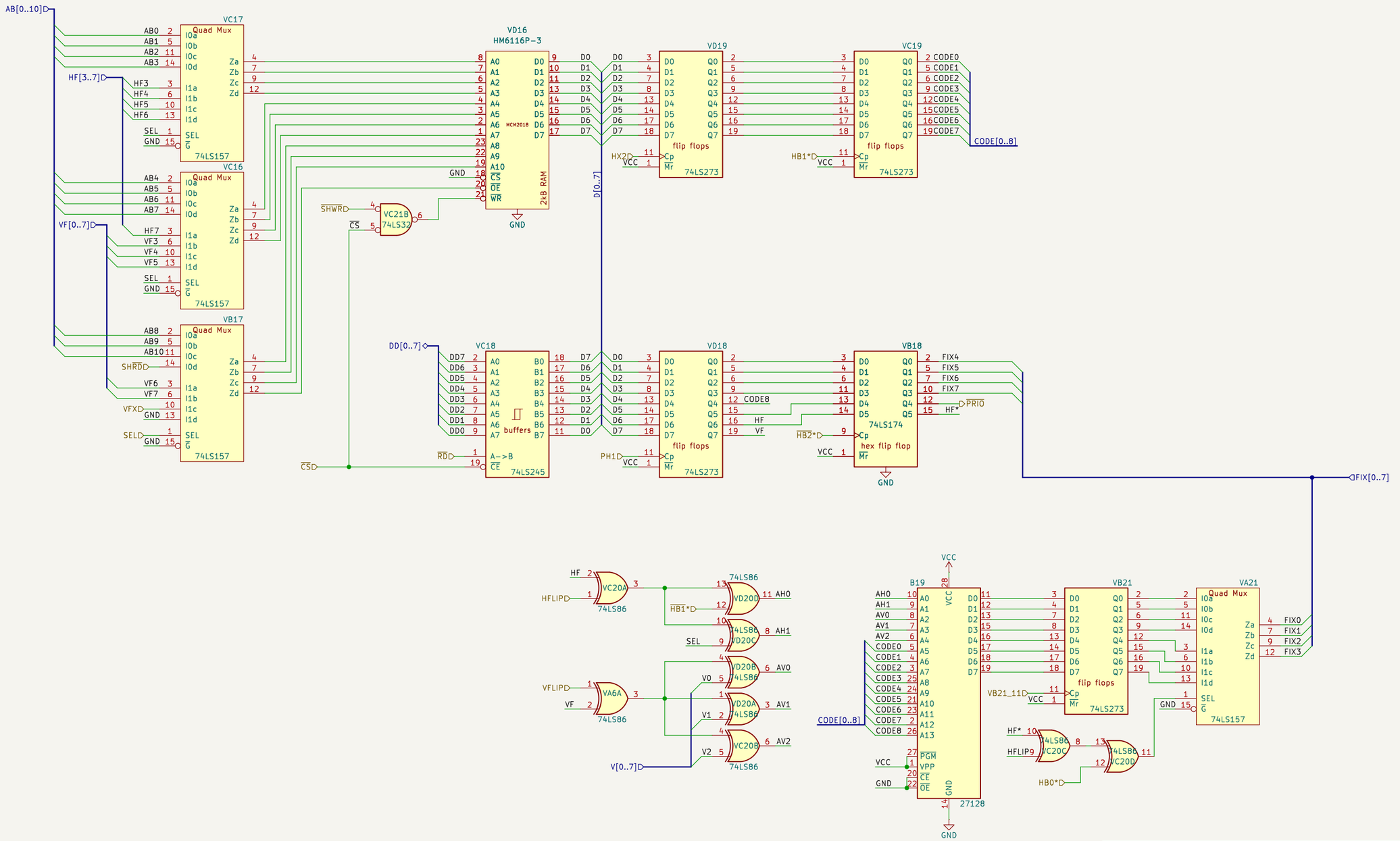


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Sheet: /video board/scroll/
File: scroll.kicad_sch

Title: Scroll Subsystem

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 12/26

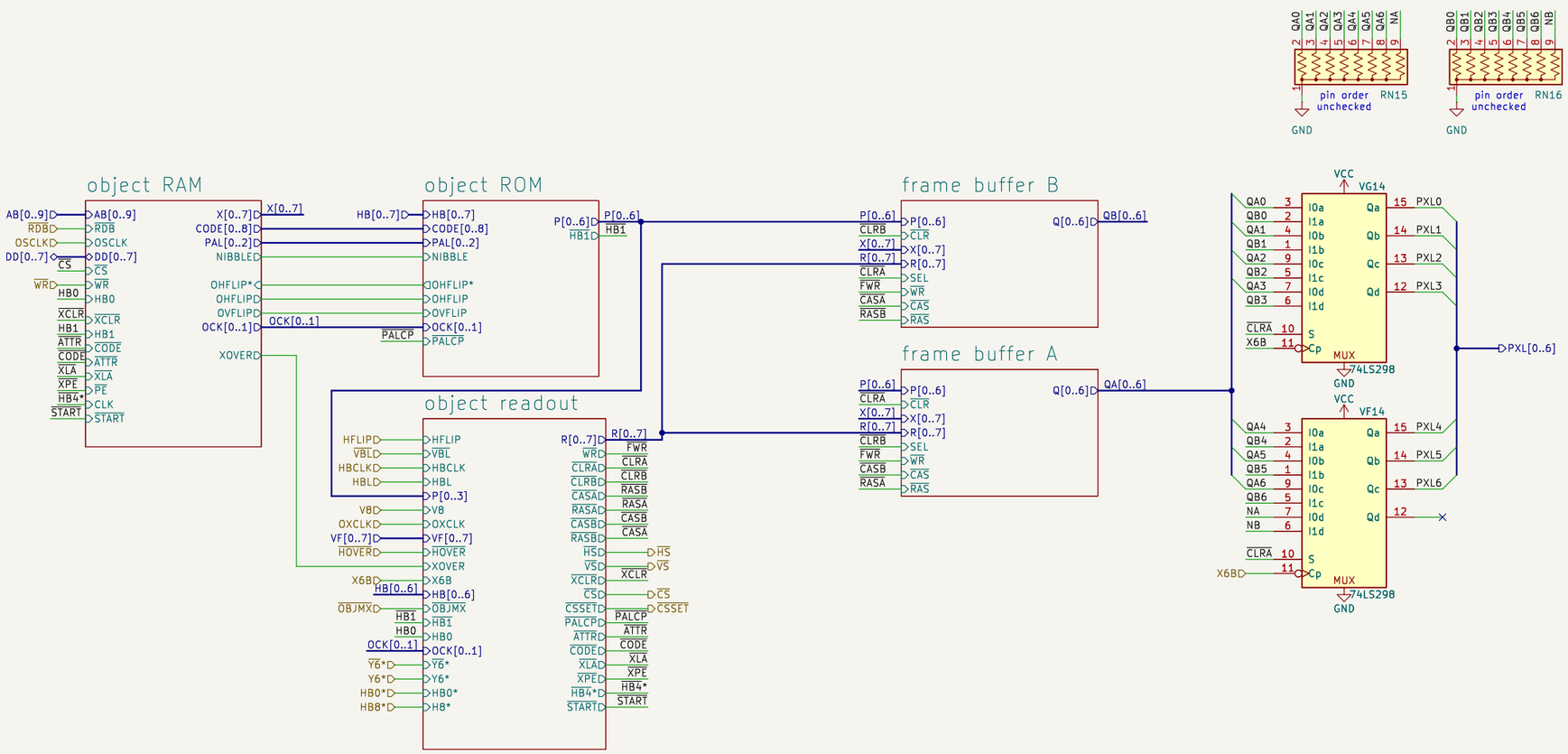


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Sheet: /video board/fix/
File: fix.kicad_sch

Title: Fixed Character Layer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 13/26



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Sheet: /video board/object/
File: object.kicad_sch

Title: Object Subsystem

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 14/26

(3.57MHz on bootleg)

Y3
4.0MHz
101
101
GND

VI22A 4069
VI22B 4069
VH22A 74LS04
CLK4

clock divider by 3

VI20B 74LS107
VI20A 74LS107
Y6
X6
HCLK
6.144 MHz
VCC

K15
330
VJ15B
74LS04
C3
101
Y1
18.432
(18.000 on bootleg)
330
VJ15A
74LS04

VI16 74LS161
VI15 74LS161
GND
VCC
D0 D1 D2 D3
Q0 Q1 Q2 Q3
PE
CEP
CET
CP
MR
TC
VCLK
counter
LSB
MSB
V[0..8]
V0
VTC
VG20F
74LS04

VI18 74LS161
VI19 74LS161
H0 H1 H2 H3
H4 H5 H6 H7 H8
TCH

VG19A 74LS74
HCLK
H0
3MHz

74LS00
VH20B
VG20C
VG20B
H8
H6
H4

VG21A 74LS74
HOVER
VCLK

VG19B 74LS74

74LS86
VH16D
VH16C
VH16B
VH16A
VF0 VF1 VF2 VF3
V1 V2 V3 V4 V5 V6 V7
VH15D
VH15C
VH15B
VH15A
VF4 VF5 VF6 VF7

VG18B
VFLIP
VF[0..7]

74LS20
VG17B
VG18C
VG17A
VH20D
X6B
TCH
HB6
HB7
HB5
VFX
HB1*
object scan clock
OSCLK
HBL
74LS27
VG16A
74LS08
74LS20
74LS00
HBCLK
HB0*
VG16B
VE21B
SSDSET
VG20A
OXCLK

VH17
3-StateHexBuffers
74LS367
HB[0..7]
HB0 HB1 HB2 HB3 HB4 HB5 HB6 HB7
X6
HCLK
Y6
Y6*
X6B
HBCLK
HB6
HB7

74LS32
VC22B
HB3
H8*
VD21A
74LS74
C22_13
101

VF21
74LS175
FOUR FLIP FLOP WITH DOUBLE RAIL OUTPUTS
HB0* VB21_11 HB1* HB1* VFX HB2*

74LS08
VF20C
VH20C
VG21B
VBL
VBL*

74LS74
VF22A
VFX
HB1*
PH1
PH1*
VG22A
VG22B
HB011
VH22E
VH22F
VF20D
VBL
HBL*
DISP

VE20C
74LS32
HBCLK
GND
PXLCLK

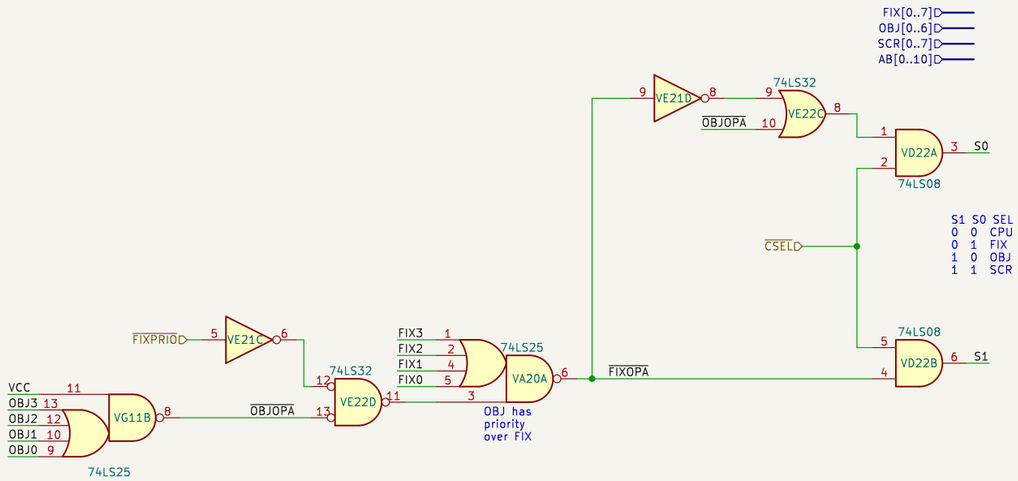
VCC
VDD
VI22G 4069
VSS
GND
VE21G 74LS04
VG17C 74LS20
VG16D 74LS27
VF20E 74LS08
VE20E 74LS32
VE16C 74LS74
VG19C 74LS74
VJ15G 74LS04
VG18E 74LS08
VG15D 74LS10
VH20E 74LS00
VH18E 74LS86
VH19E 74LS86
VH16E 74LS86
VI20C 74LS107
VF22C 74LS74
VG22C 74LS74
VG20G 74LS04
VG21C 74LS74
VH22G 74LS04

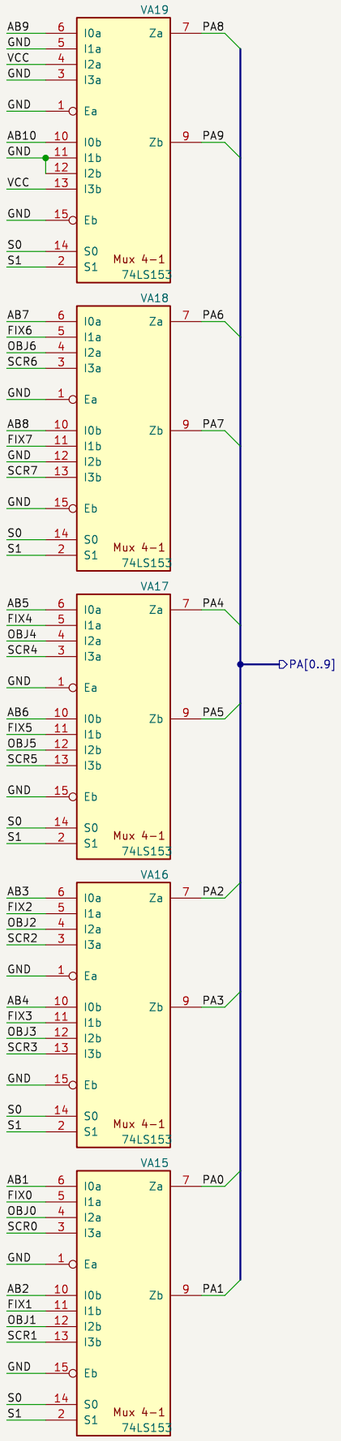
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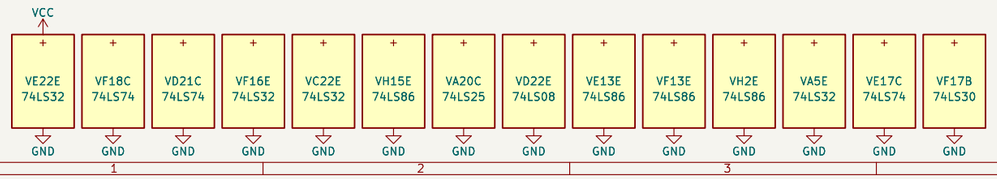
Sheet: /video board/timing/
File: timing.kicad_sch

Title: Video Timing

Size: A3 | Date: 2024-11-15 | Rev:
KiCad E.D.A. 8.0.9 | Id: 15/26





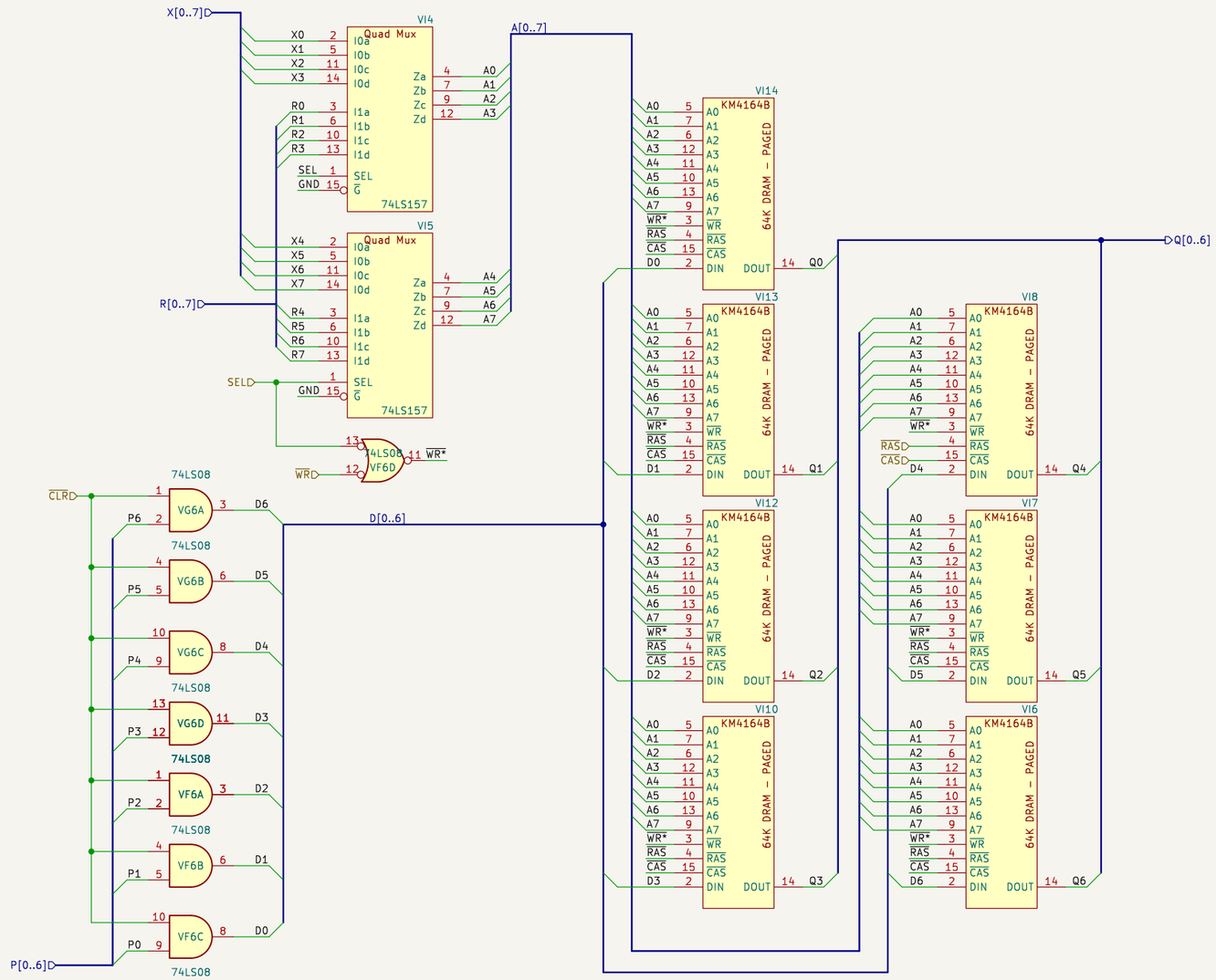


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Sheet: /video board/color mixer/
File: colmix.kicad_sch

Title: Color mixer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 16/26

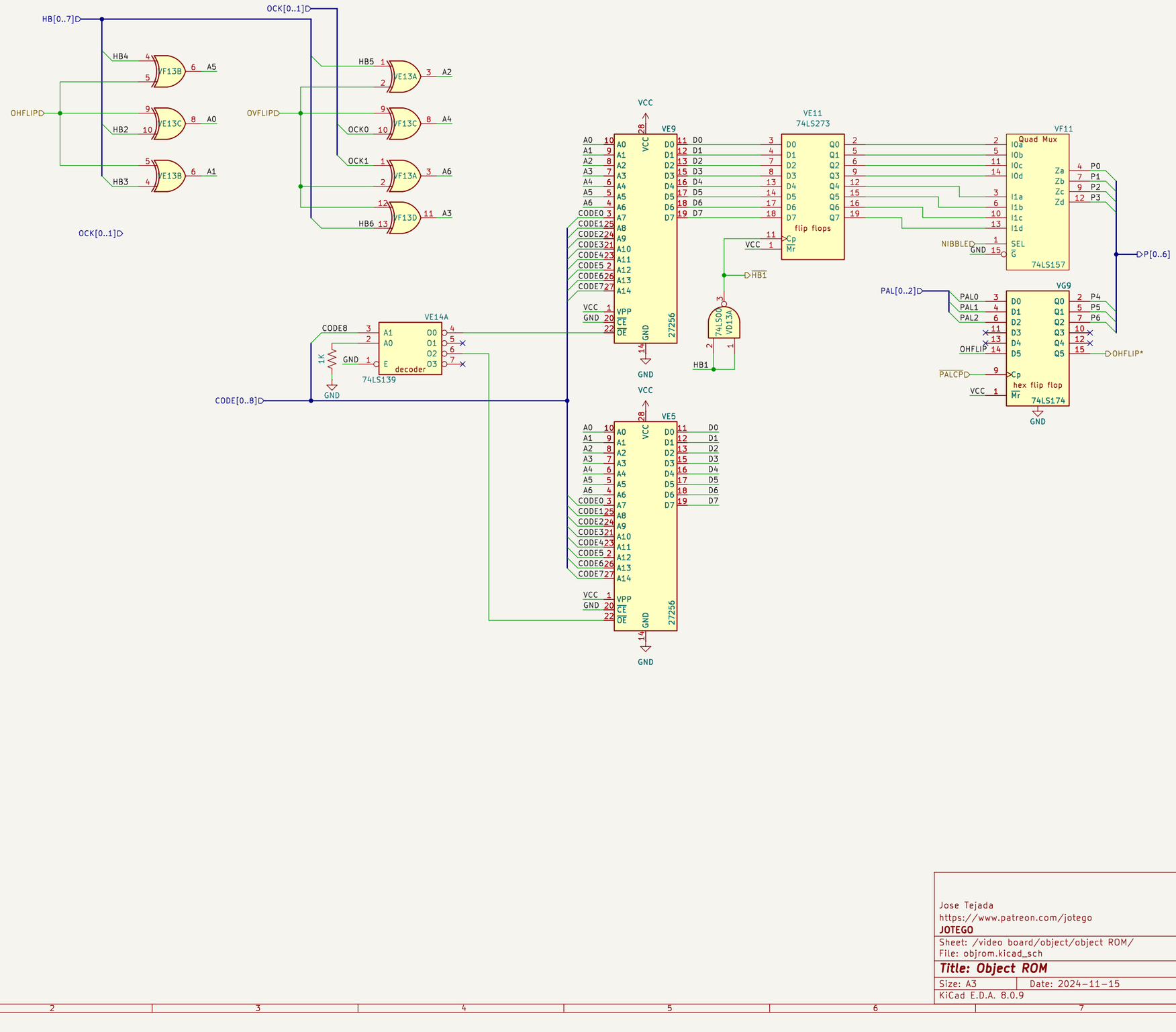


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Sheet: /video board/object/frame buffer A/
File: buffer.kicad_sch

Title: Line Buffer

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 17/26

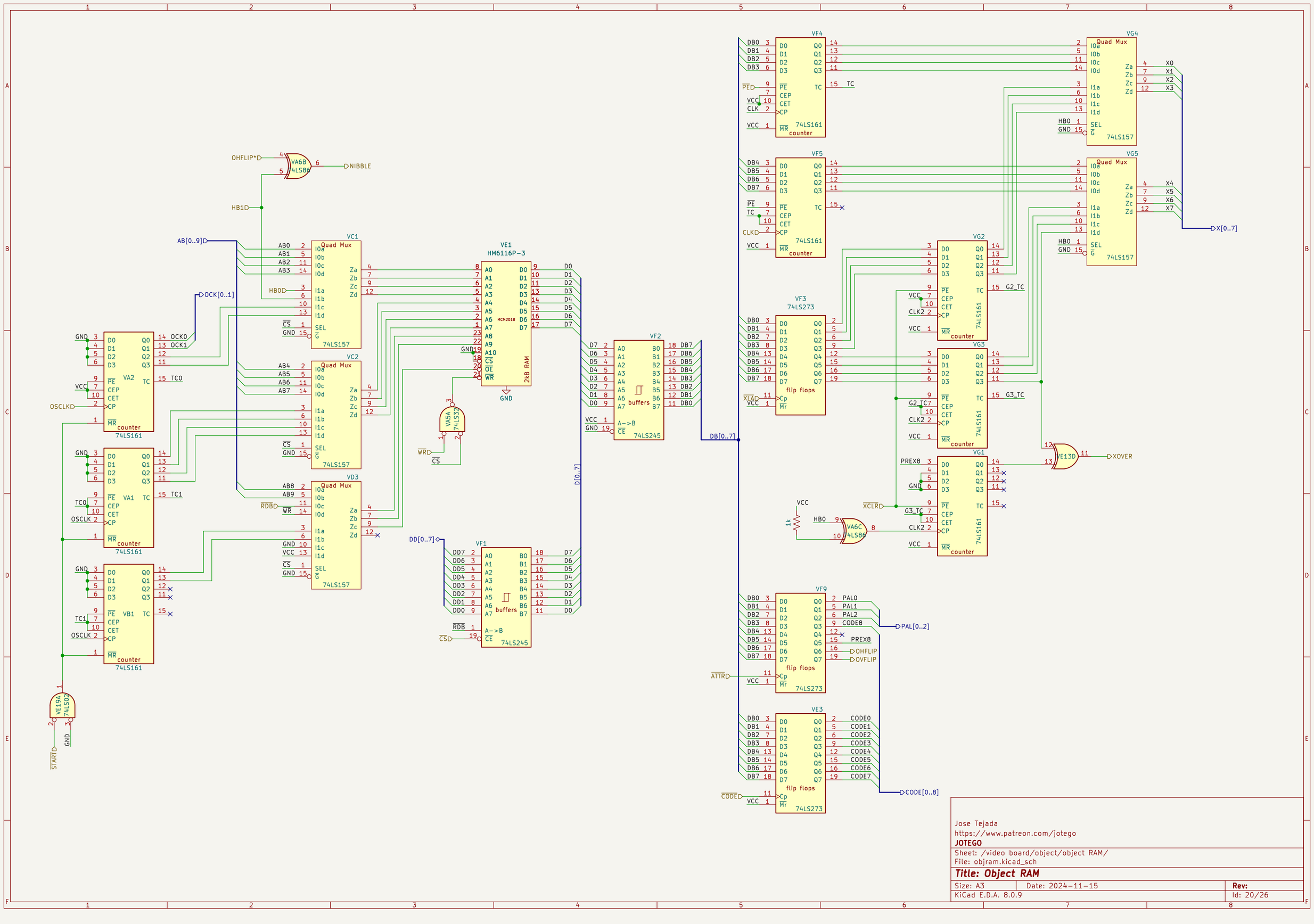


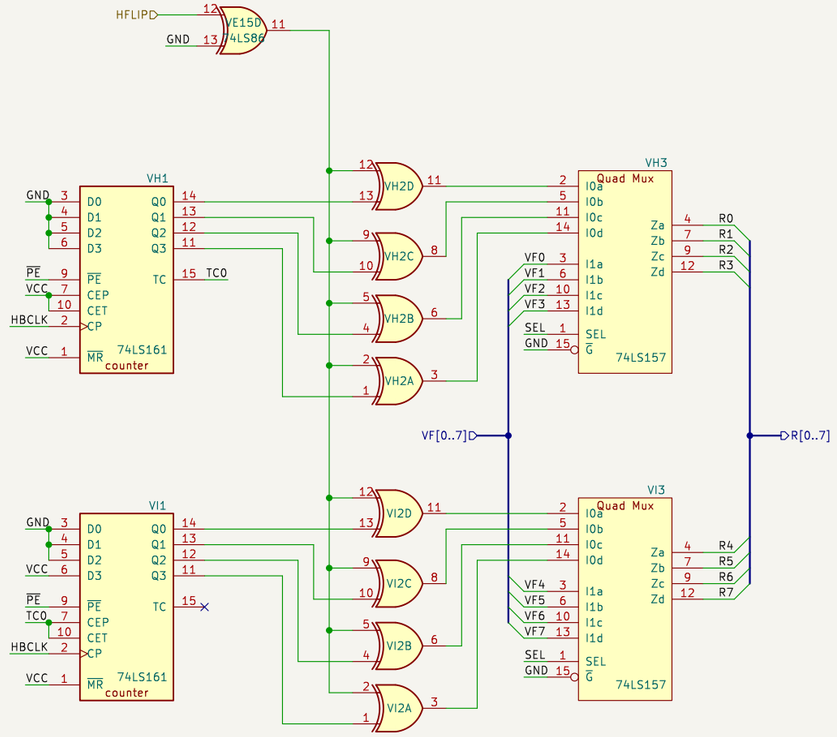
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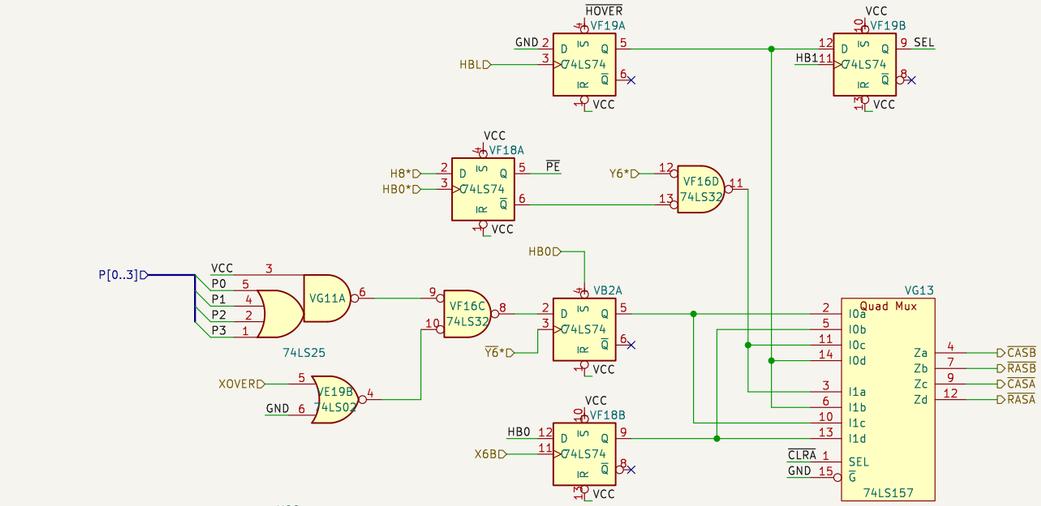
Sheet: /video board/object/object ROM/
File: objrom.kicad_sch

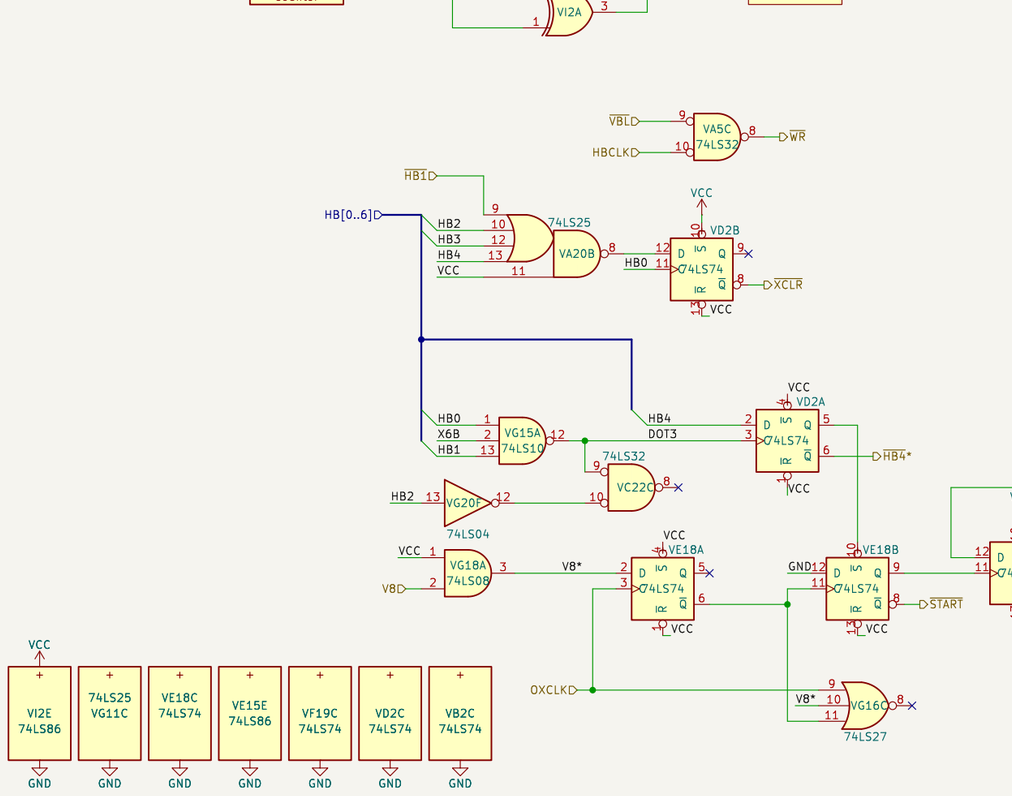
Title: Object ROM

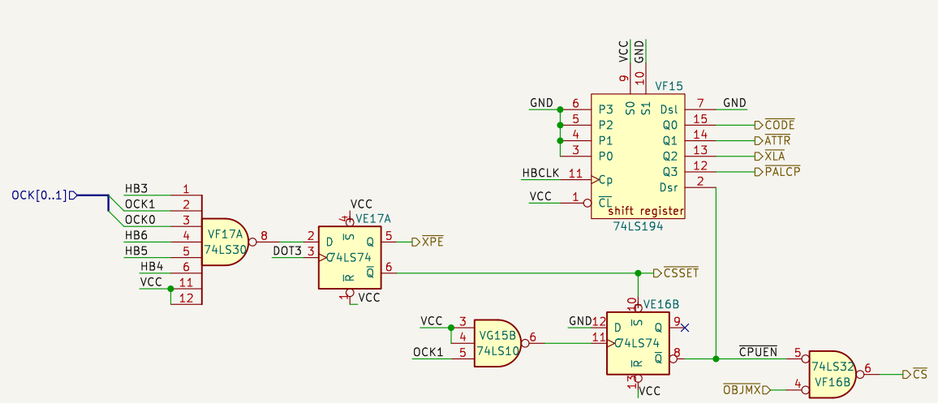
Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 19/26


OHFLIP*
VA6B
74LS86
NIBBLE
HB1
AB[0..9]
OCK[0..1]
VC1
Quad Mux
AB0
AB1
AB2
AB3
HB0
Za
Zb
Zc
Zd
74LS157
VE1
HM6116P-3
2kB RAM
GND
D0
D1
D2
D3
D4
D5
D6
D7
VF2
buffers
A->B
74LS245
DB[0..7]
VF4
VF5
VF3
74LS273
flip flops
VG4
VG5
VG2
VG3
VG1
74LS161
counter
VA2
VA1
VB1
VC2
VD3
VF1
VE19A
74LS02
VE13D
VA5A
VA6C
VE3
VF9
OSCLK
START
XOVER
X[0..7]
PAL[0..2]
CODE[0..8]
OHFLIP
OVFLIP
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Sheet: /video board/object/object RAM/
File: objram.kicad_sch
Title: Object RAM
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 8.0.9
Id: 20/26







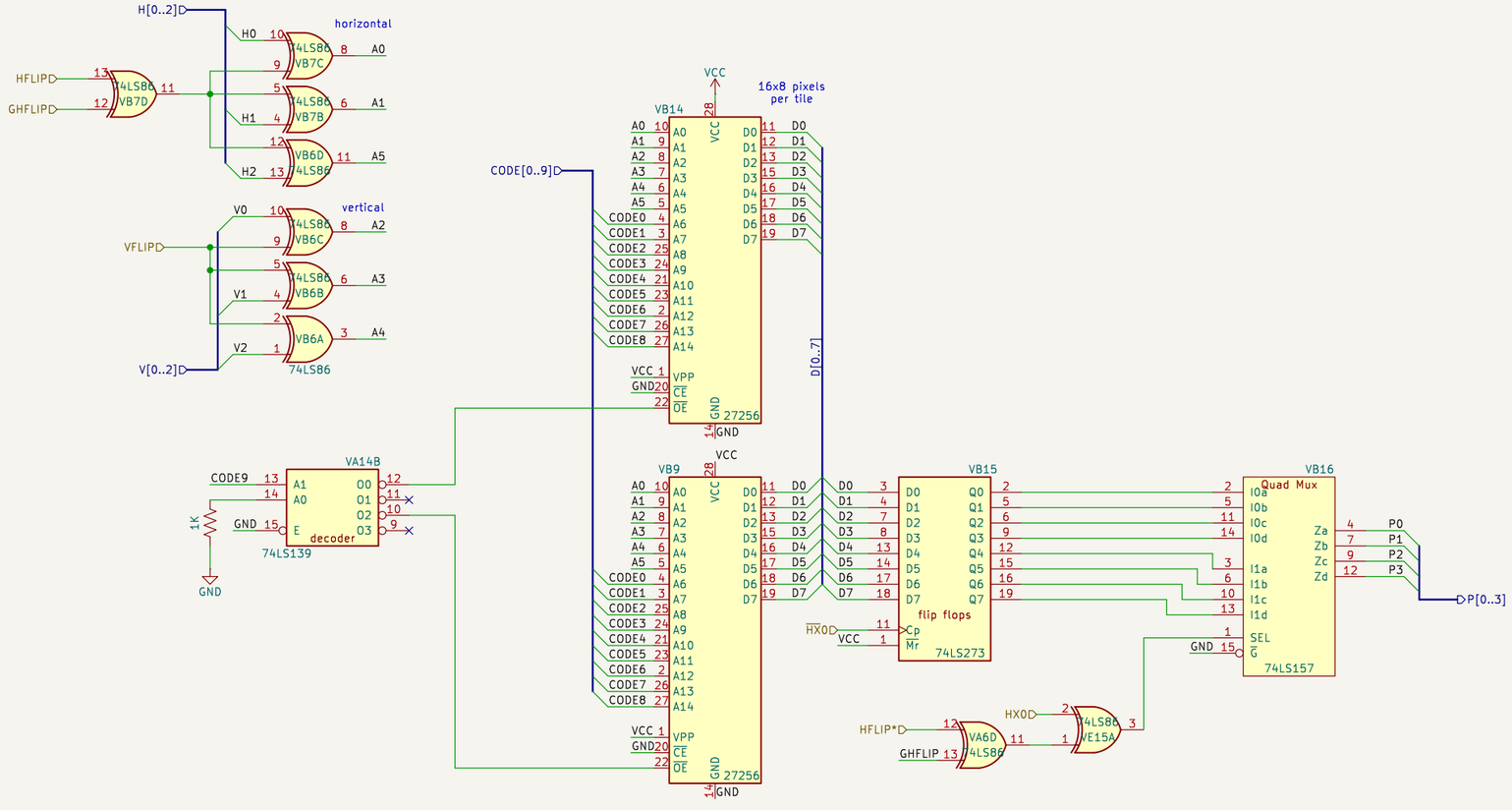


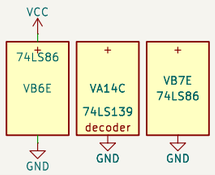
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Sheet: /video board/object/object readout/
File: objread.kicad_sch

Title: Object readout sequencer

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 21/26



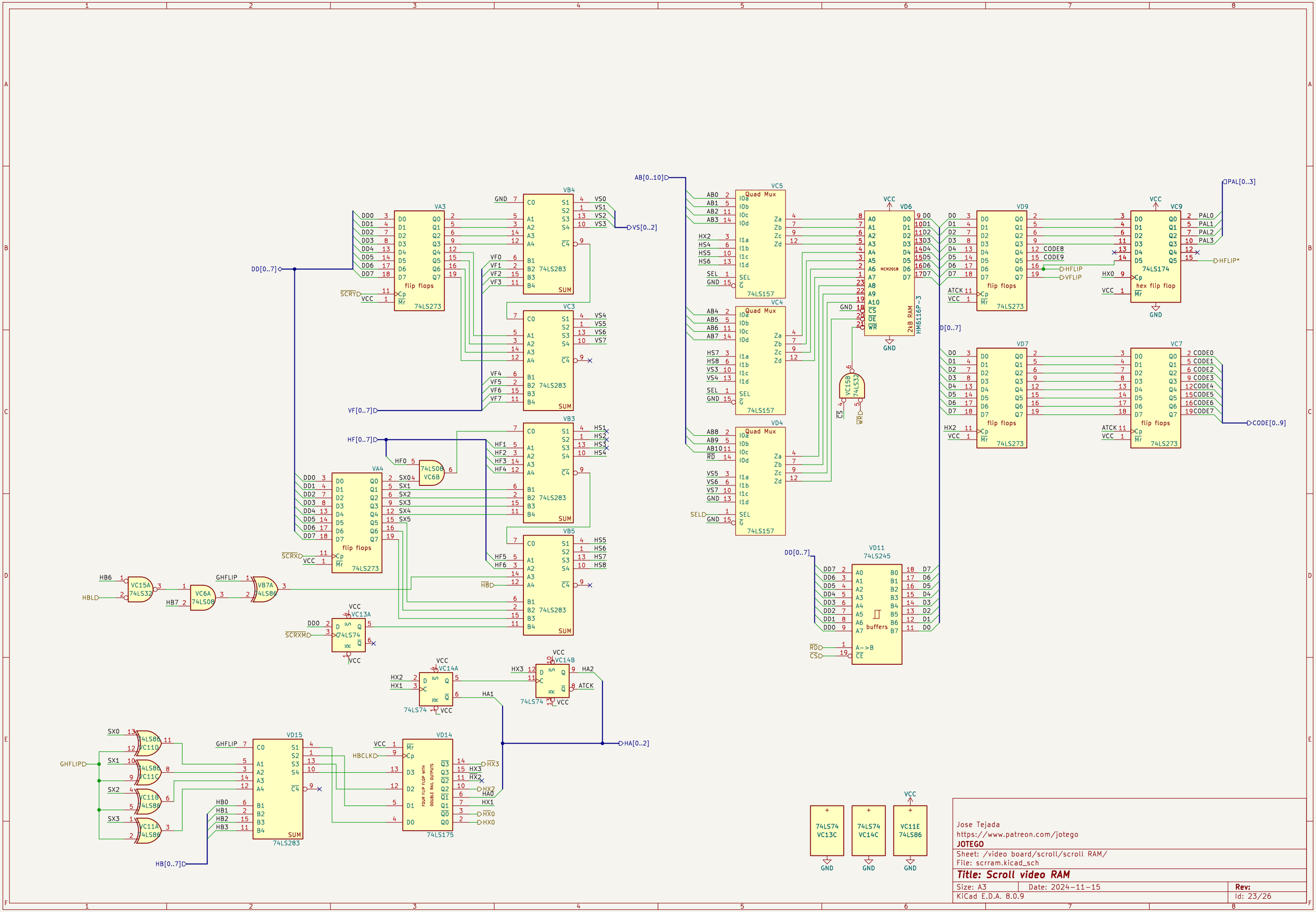


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Sheet: /video board/scroll/scroll ROM/
File: scrrom.kicad_sch

Title: scroll ROM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 22/26

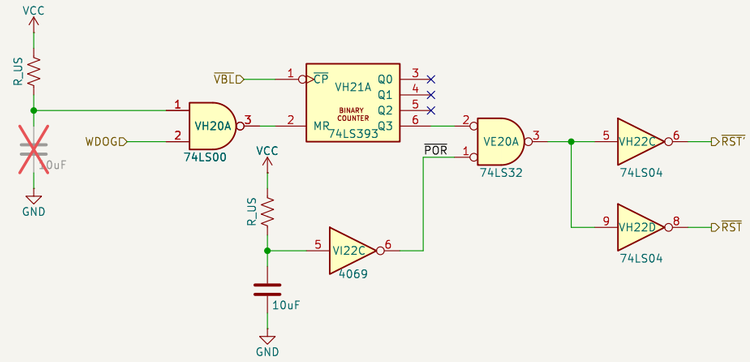


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Sheet: /video board/scroll/scroll RAM/
File: scrram.kicad_sch

Title: Scroll video RAM

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 23/26



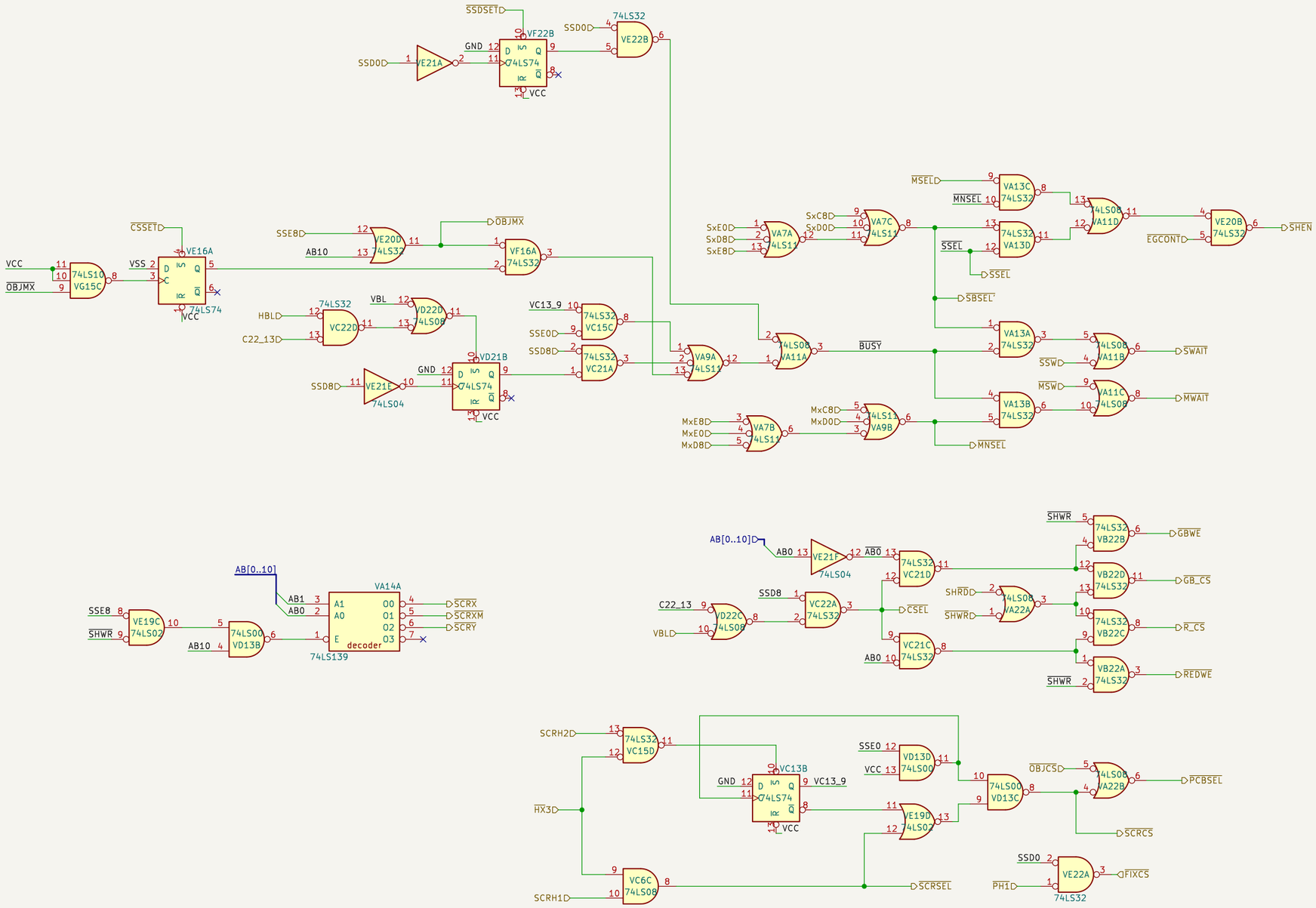


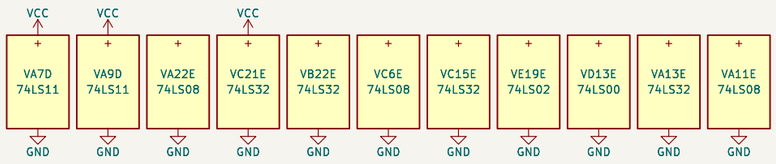
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Sheet: /video board/watchdog/
File: watchdog.kicad_sch

Title: Watchdog Circuit

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 24/26



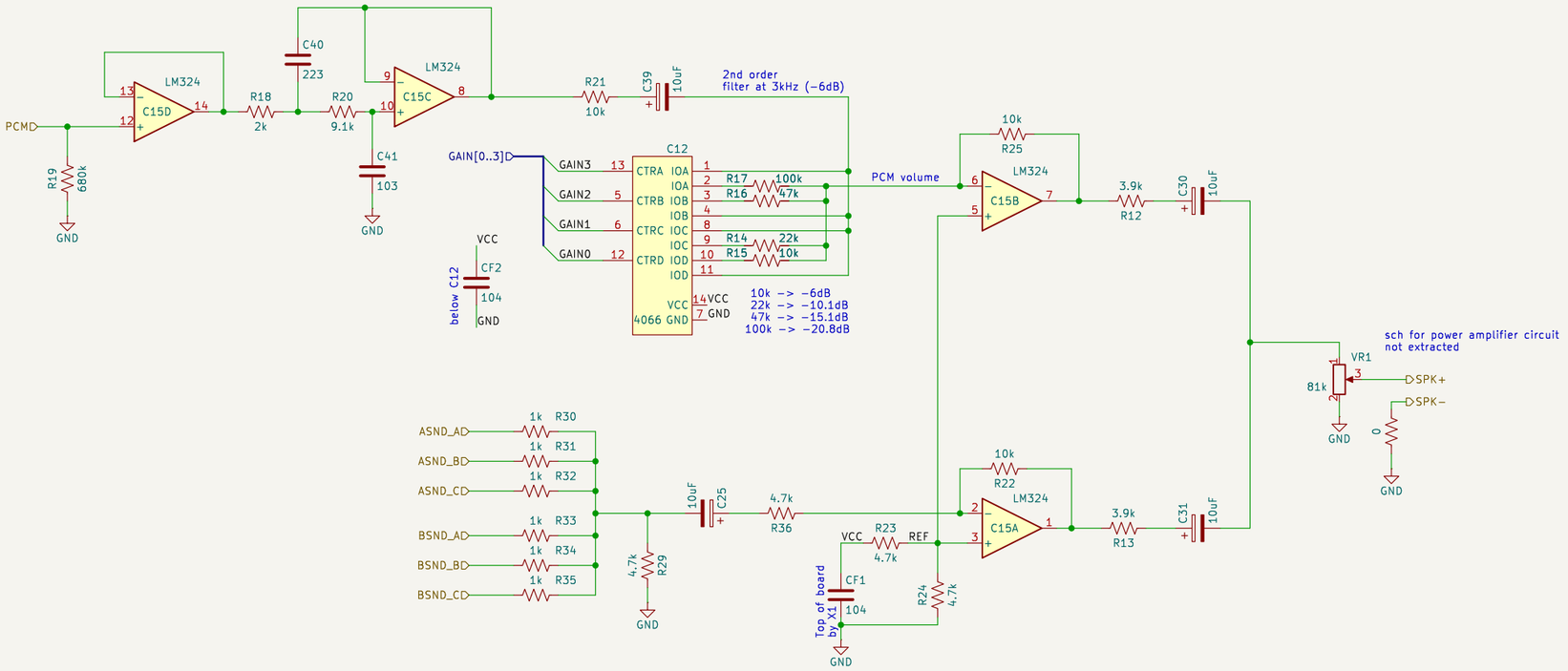


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Sheet: /video board/address decoder/
File: decoder.kicad_sch

Title: Address decoder

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 25/26



Passive component references
when shown, came from original PCB.
The bootleg did not have
references for passives.

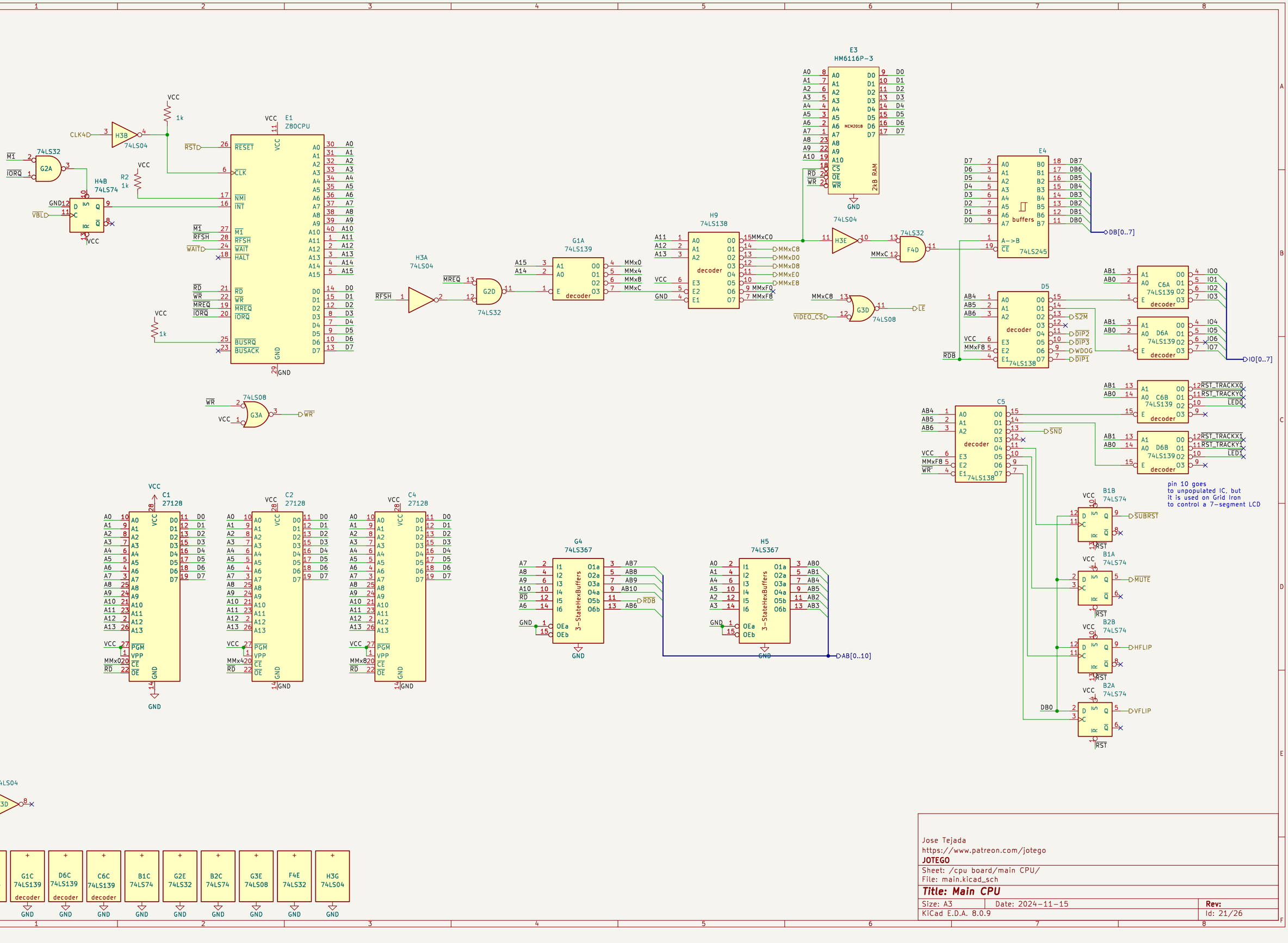
Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

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Sheet: /cpu board/sound/analog/
File: analog.kicad_sch

Title: Sound amplifiers

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 26/26



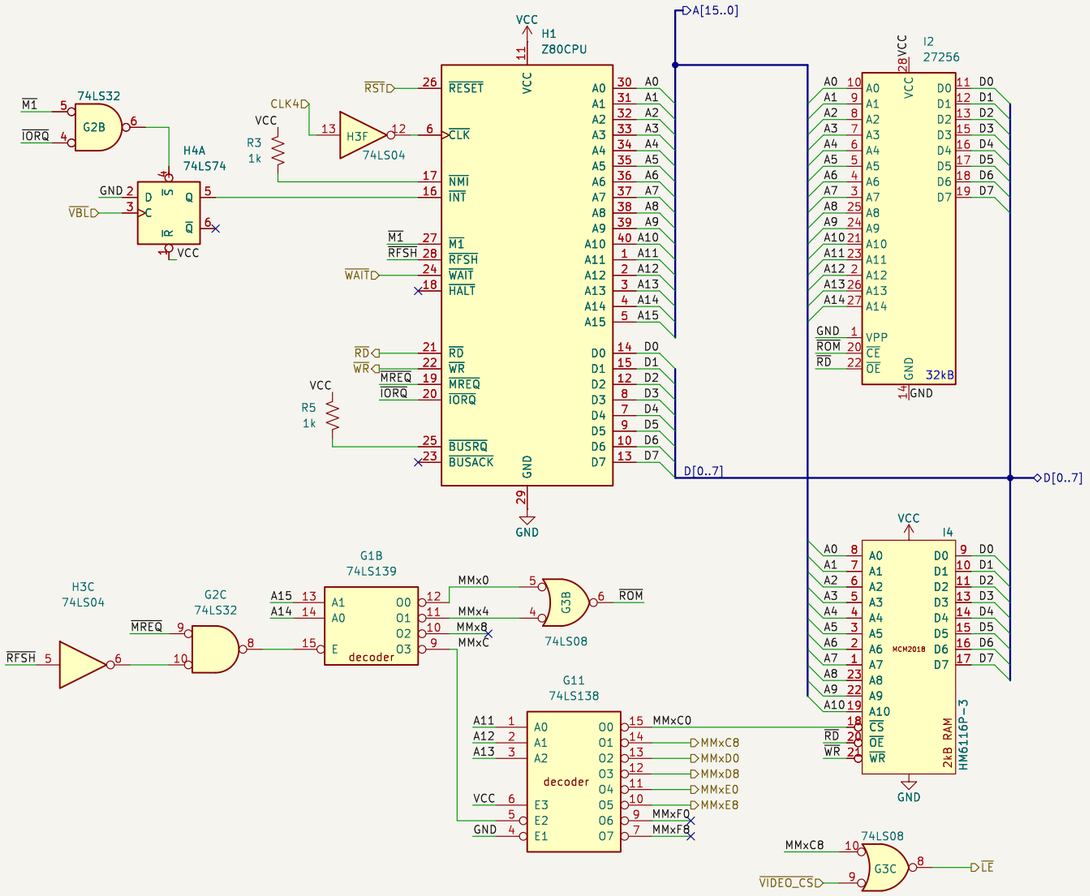
Jose Tejada
https://www.patreon.com/jotego
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Sheet: /cpu board/main CPU/
File: main.kicad_sch

Title: Main CPU

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 21/26

H16B
74LS139
A15 13
A14 14
A1
A0
O0 12 MMx0
O1 11 MMx4
O2 10 MMx8
O3 9 MMxC
E
decoder
H17A
74LS04
RFSH 1
MREQ 1
74LS32
G17A
3
15
H16A
74LS139
AB1 3
AB0 2
O0 4 PCMSET
O1 5 PCMCTL
O2 6 GAINSET
O3 7 NMICLR
E17B
74LS32
MMx8 4
WRB 5
6
1
F12
YM2149
VCC
D0 37
D1 36
D2 35
D3 34
D4 33
D5 32
D6 31
D7 30
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
VCC 25
A8
GND 24
A9
ABDIR 27
BDIR
ABC1 29
BC1
ABC2 28
BC2
PSGCLK 22
CLOCK
MUTE 23
SEL
RESET
39 TEST1
VSS
1 GND
40 VCC
A 3 ASND_A
B 4 ASND_B
C 38 ASND_C
21
IOA0 20 CNT0
IOA1 19 CNT1
IOA2 18 CNT2
IOA3 17 CNT3
IOA4 16 CNT4
IOA5 15 CNT5
IOA6 14 CNT6
IOA7
IOB0 13 CNT7
IOB1 12 CNT8
IOB2 11 CNT9
IOB3 10 CNT10
IOB4 9 CNT11
IOB5 8 CNT12
IOB6 7 CNT13
IOB7 6
VCC
1k
CNT[0..13]
F15
YM2149
VCC
D0 37
D1 36
D2 35
D3 34
D4 33
D5 32
D6 31
D7 30
DA0
DA1
DA2
DA3
DA4
DA5
DA6
DA7
VCC 25
A8
GND 24
A9
BBDIR 27
BDIR
BBC1 29
BC1
BBC2 28
BC2
PSGCLK 22
CLOCK
MUTE 23
SEL
RESET
39 TEST1
VSS
1 GND
40 VCC
A 3 BSND_A
B 4 BSND_B
C 38 BSND_C
21 SMP0
IOA0 20 SMP1
IOA1 19 SMP2
IOA2 18 SMP3
IOA3 17 SMP4
IOA4 16 SMP5
IOA5 15 SMP6
IOA6 14 SMP7
IOA7
IOB0 13 SMP8
IOB1 12 SMP9
IOB2 11 SMP10
IOB3 10 SMP11
IOB4 9 SMP12
IOB5 8 SMP13
IOB6 7 SMP14
IOB7 6 SMP15
SMP[0..15]
VCC
Taps to unpopulated sockets
NMICLR
E2C
74LS08
G6B
74LS74
GND 12
SND 11
D
C
S
R
Q
10
9
8
13
VCC
M1
IORQ
G17C
74LS32
I16A
74LS74
GND 2
SIRQ 3
5
6
1
VCC
G13
Z80CPU
MUTE 26
CPUCLK 6
NMI 17
INT 16
M1 27
RFSH 28
WAIT 24
HALT 18
RD 21
WR 22
MREQ 19
IORQ 20
BUSRQ 25
BUSACK 23
RESET
CLK
VCC 11
GND 29
A0 30
A1 31
A2 32
A3 33
A4 34
A5 35
A6 36
A7 37
A8 38
A9 39
A10 40
A11 1
A12 2
A13 3
A14 4
A15 5
D0 14
D1 15
D2 12
D3 8
D4 7
D5 9
D6 10
D7 13
1K
1K
VCC
H13
27128
VCC 28
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
PGM 27
VPP 1
CE 20 MMx0
OE 22 RDB
GND 14
D0 11
D1 12
D2 13
D3 15
D4 16
D5 17
D6 18
D7 19
GND
I14
HM6116P-3
A0 8
A1 7
A2 6
A3 5
A4 4
A5 3
A6 2
A7 1
A8 23
A9 22
A10 19
MMx8 18
RDB 20
WRB 21
CS
OE
WR
MCM2018
2kB RAM
D0 9
D1 10
D2 11
D3 13
D4 14
D5 15
D6 16
D7 17
D[0..7]
74LS04
H17C
CLK
6
5
I16B
74LS74
VCC
12
11
10
13
D
C
S
R
Q
9 PSGCLK
8
74LS04
H17D
9
8 CPUCLK
MD[0..7]
E11
MD0 3
MD1 4
MD2 7
MD3 8
MD4 13
MD5 14
MD6 17
MD7 18
D0
D1
D2
D3
D4
D5
D6
D7
3-state
flip flops
OE
74LS374
2 DB0
5 DB1
6 DB2
9 DB3
12 DB4
15 DB5
16 DB6
19 DB7
E12
DB0 3
DB1 4
DB2 7
DB3 8
DB4 13
DB5 14
DB6 17
DB7 18
2 MD0
5 MD1
6 MD2
9 MD3
12 MD4
15 MD5
16 MD6
19 MD7
F17
74LS367
3-StateHexBuffers
RD 2
A0 4
6
A1 10
WR 12
IORQ 14
GND 1
15
OEa
OEb
I1
I2
I3
I4
I5
I6
O1a 3 RDB
O2a 5 ABO
O3a 7
O4a 9 AB1
O5b 11 WRB
O6b 13 IORQB
MMxC 2
F4A
74LS32
1
3
SND 11
WRB 12
74LS32
G17D
MMxC 13
11
S2M
PCM
PCMCTL
PCMSET
DBO
MUTE
SMP[0..15]
CNT[0..13]
CTL
SET
DBO
MUTE
SMP[0..15]
DA[1..14]
PCMSND
PCMSND
analog
ASND_A
ASND_B
ASND_C
BSND_A
BSND_B
BSND_C
GAIN[0..3]
PCMSND
ASND_A
ASND_B
ASND_C
BSND_A
BSND_B
BSND_C
GAIN[0..3]
PCM
SPK+
SPK-
SPK+
SPK-
File: analog.kicad_sch
F11
74LS245
DO 2
D1 3
D2 4
D3 5
D4 6
D5 7
D6 8
D7 9
A0
A1
A2
A3
A4
A5
A6
A7
buffers
A->B
RDB 1
19
CE
B0 18 DB0
B1 17 DB1
B2 16 DB2
B3 15 DB3
B4 14 DB4
B5 13 DB5
B6 12 DB6
B7 11 DB7
MMx8 12
MMxC 13
E10D
74LS08
11
D12
DB1 3
DB2 4
6
11
DB3 13
DB0 14
D0
D1
D2
D3
D4
D5
74LS174
hex flip flop
Cp
Mr
GAINSET
MUTE 1
Q0 2 GAIN1
Q1 5 GAIN2
7
10
Q3 12 GAIN3
Q4 15 GAIN0
Q5
GAIN[0..3]
G6A
74LS74
D
C
S
R
Q
2
3
5
6
74LS02
G16C
8
9
10
VCC
H17G 74LS04
H16C 74LS139 decoder
E10E 74LS08
I16C 74LS74
F16E 74LS02
E2E 74LS08
G16E 74LS02
G6C 74LS74
GND
74LS02
F16B
WRB 6
5
4 ABDIR
74LS02
F16D
ABO 11
12
13 ABC2
74LS02
F16A
RDB 2
3
1 ABC1
74LS02
G16B
WRB 6
5
4 BBDIR
74LS02
G16D
ABO 11
12
13 BBC2
74LS02
G16A
RDB 2
3
1 BBC1
IORQB 12
AB1 13
H17F
74LS04
12
13
E17D
74LS32
11 IOA1
74LS32
E17C
10
9
8 IOA1
BDIR BC2 BC1
0 X 0 inactive
1 0 0 latch address
1 1 0 data write
0 0 1 latch address
0 1 1 data read
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JOTEGO
Sheet: /cpu board/sound/
File: sound.kicad_sch
Title: Sound
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 8.0.9
Id: 22/26



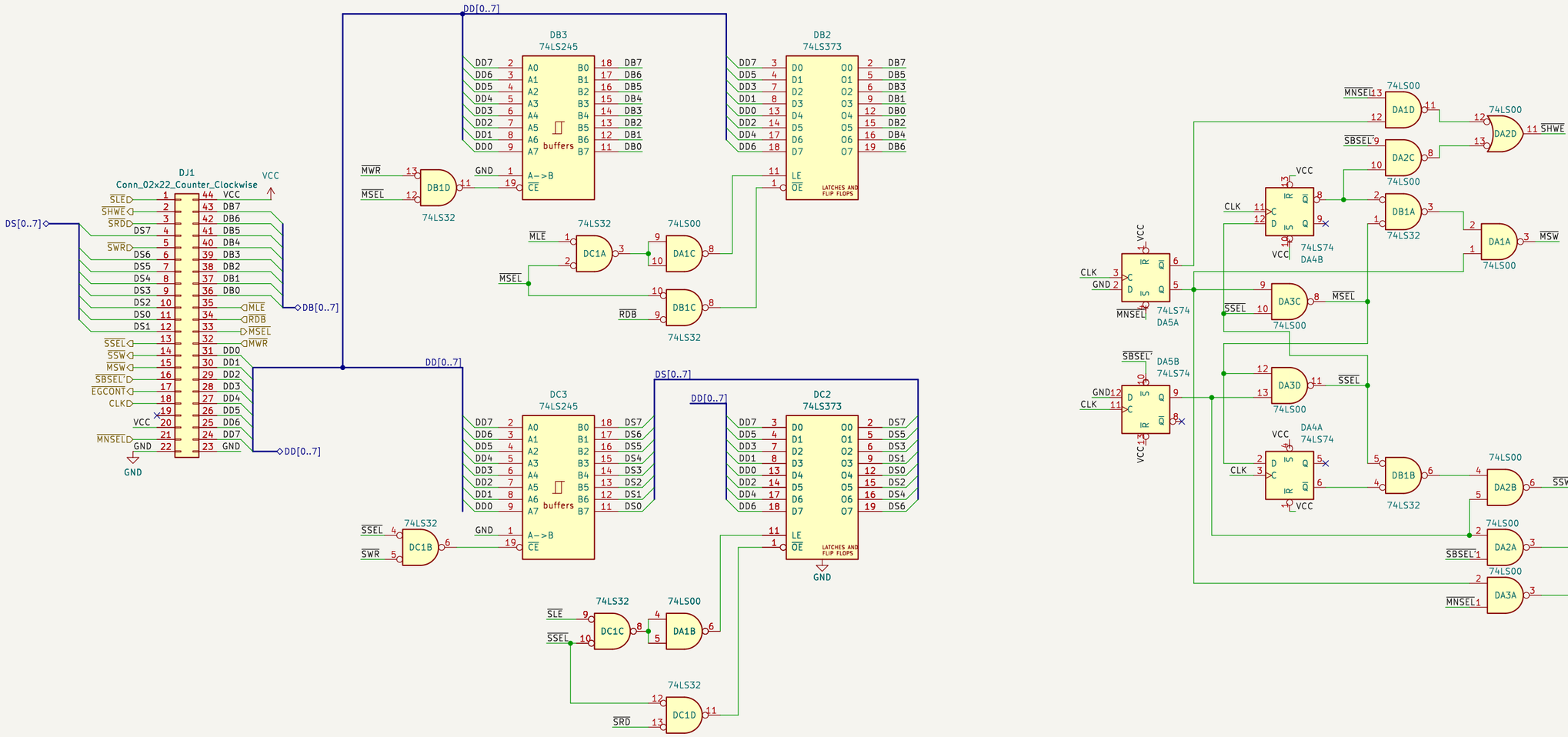
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Sheet: /cpu board/sub CPU/
File: subcpu.kicad_sch

Title: Subordinate CPU

Size: A3	Date: 2024-11-15	**Rev:**
KiCad E.D.A. 8.0.9		Id: 23/26

F5
SW_DIP_x08
E6
74LS244
GND
E5
74LS244
SW_DIP_x08
F6
B3
3-StateHexBuffers
74LS367
A2
SW_DIP_x04
DIP2
DIP3
DIP1
DB[0..7]
VCC
RN17
R_Network04
COIN[1..2]
COIN1
COIN2
START1
START2
START[1..2]
100
C
B12
IO[0..7]
IO2
Pull-up pin order
may not match the PCB
RN13
RPU15
RPU14
RPU13
RPU12
RPU11
RPU10
RPU9
RPU8
RN19
RPU23
RPU22
RPU21
RPU20
RPU19
RPU18
RPU17
RPU16
B11
IO3
SHOT1
100RD2
SHOT2
100RD1
B4
IO7
SHOT[1..2]
IO side of the board
contains unpopulated
footprints. The schematics
do not reflect the
full PCB on this side.
220
RN18
R_Network08_US
PU0
PU1
PU2
PU3
PU4
PU5
PU6
PU7
DOWN[1..2]
UP[1..2]
RIGHT[1..2]
LEFT[1..2]
maybe through
some pullup?
LEFT1
ROD1
B5E
74LS04
RIGHT1
ROC1
UP1
ROF1
B5D
DOWN1
ROE1
LEFT2
ROB1
B7B
RIGHT2
ROA1
UP2
ROH1
B7C
DOWN2
ROG1
C10
Quad Mux
74LS157
C9
C8
C7
D10
D9
D8
D7
IO0
IO1
IO4
IO5
IO[0..7]
B5G
B7G
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Sheet: /cpu board/IO/
File: io.kicad_sch
Title: Cabinet Input/Output
Size: A3
Date: 2024-11-15
Rev:
KiCad E.D.A. 8.0.9
Id: 24/26



This circuit takes three small boards on the original PCB.
These boards were manufactured by Mitsubishi and contain the circuits seen on the bootleg
– MA7053 x2, 28 pin each
– MA7054 x1, 15 pins
The layout around the daughter board on the bootleg differs slightly from the original in order to adjust the pinout to a single 44–pin board.

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Sheet: /cpu board/daughter/
File: daughter.kicad_sch

Title: Small daughter board

Size: A3	Date: 2024–11–15	Rev:
KiCad E.D.A. 8.0.9		Id: 25/26

Resistor networks reference:
RN1 = E9 top
RN2 = below RN1
...
RN11 = above RN12
RN12 = E7 bottom

PA[0..9]

VCC

G9
4096 BIT STATIC RAM MN2114
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9
REDWE
GND
D0 D1 D2 D3

F9
74LS273 flip flops
B3 B2 B1 B0 B0 B1 B2 B3
PXLCLK
RST

2.2k
RN4 R from pin 1 to 2 = 2.2kOhm
RN3
RN1 RED
RN2 R from pin 1 to 2 = 2.2kOhm
GND

SYMBOL DOES
NOT MATCH
INTERNAL NETWORK

F10
74LS245 buffers
R_CS
SHRD
D[0..7]
A->B
B7 B6 B5 B4 B3 B2 B1 B0

1k
GND

G10
74LS245 buffers
SHRD
GB_CS
A->B
G0 G1 G2 G3 R0 R1 R2 R3

G and R
are likely
swapped

F8
74LS273 flip flops
R3 R2 R1 R0 G3 G2 G1 G0
PXLCLK
RST

2.2k
RN8
RN7
RN6 BLUE
RN5
GND

MSB to common pin 222 Ohm
Each bit up gets 2x (verified)

VCC

G8
4096 BIT STATIC RAM MN2114
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9
GBWE
GND
D0 D1 D2 D3

F7
74LS273 flip flops
R3 R2 R1 R0 G3 G2 G1 G0
PXLCLK
RST

2.2k
RN12 GREEN
RN11
RN10
RN9
GND

VCC

G7
4096 BIT STATIC RAM MN2114
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9
GBWE
GND
D0 D1 D2 D3

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Sheet: /cpu board/palette/
File: palette.kicad_sch

Title: Color Palette

Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: #/26